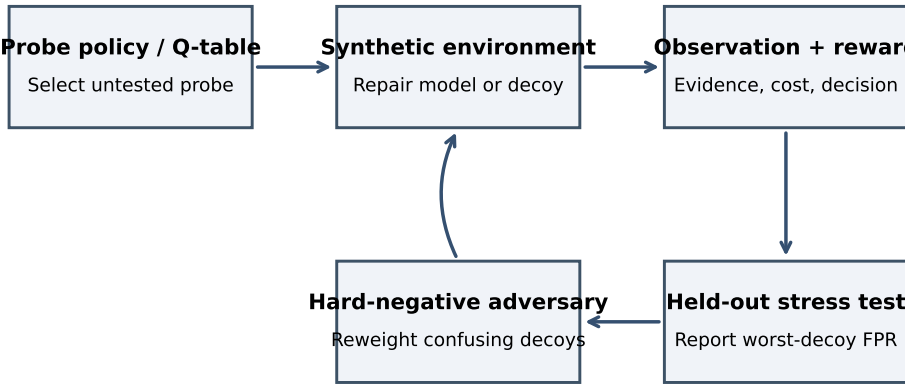Alternating probe-policy learning and hard-negative challenge sampling